VISHWAKARMA
AWARDS
A Maker Bhavan Foundation Initiative

MAKER
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VISHWAKARMA AWARDS 2026-27

VK26-IITH-SM-0275

Problem Context

GUIDELINE SLIDE

Project Overview:

Title: PRAN RAKSHAK

Team ID: **VK26-IITH-SM-0275**

Theme 5: Sustainable Mining: Worker Health, Safety & Environmental Hazard Tracking

Team Name: Codemium

Team Leader: Kaaviya Varshini V

Problem Statement: Mine workers face undetected gas and dust exposure underground, and mines discharge toxic runoff that contaminates surrounding water, because monitoring is manual, fragmented across separate systems, and unable to locate a worker when something goes wrong. We need one low-cost, offline capable system that senses gases, dust, and runoff quality, locates workers underground, and warns before limits are breached.

Problem Context

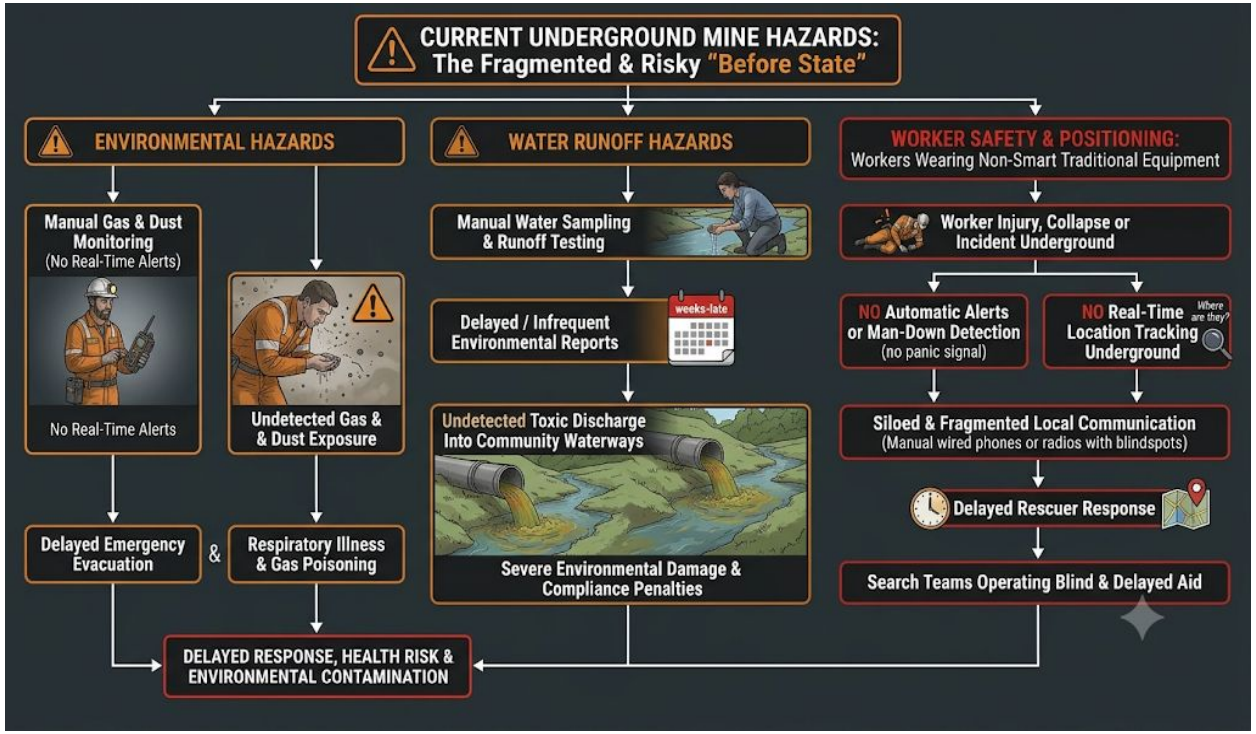
Core Innovation: An affordable, offline capable smart safety helmet that integrates real time gas & dust hazard alerts, worker localization, panic button with man down detection & direct wireless warnings to protect miners underground.



User Journey

GUIDELINE SLIDE

Visual "Before State" Requirement



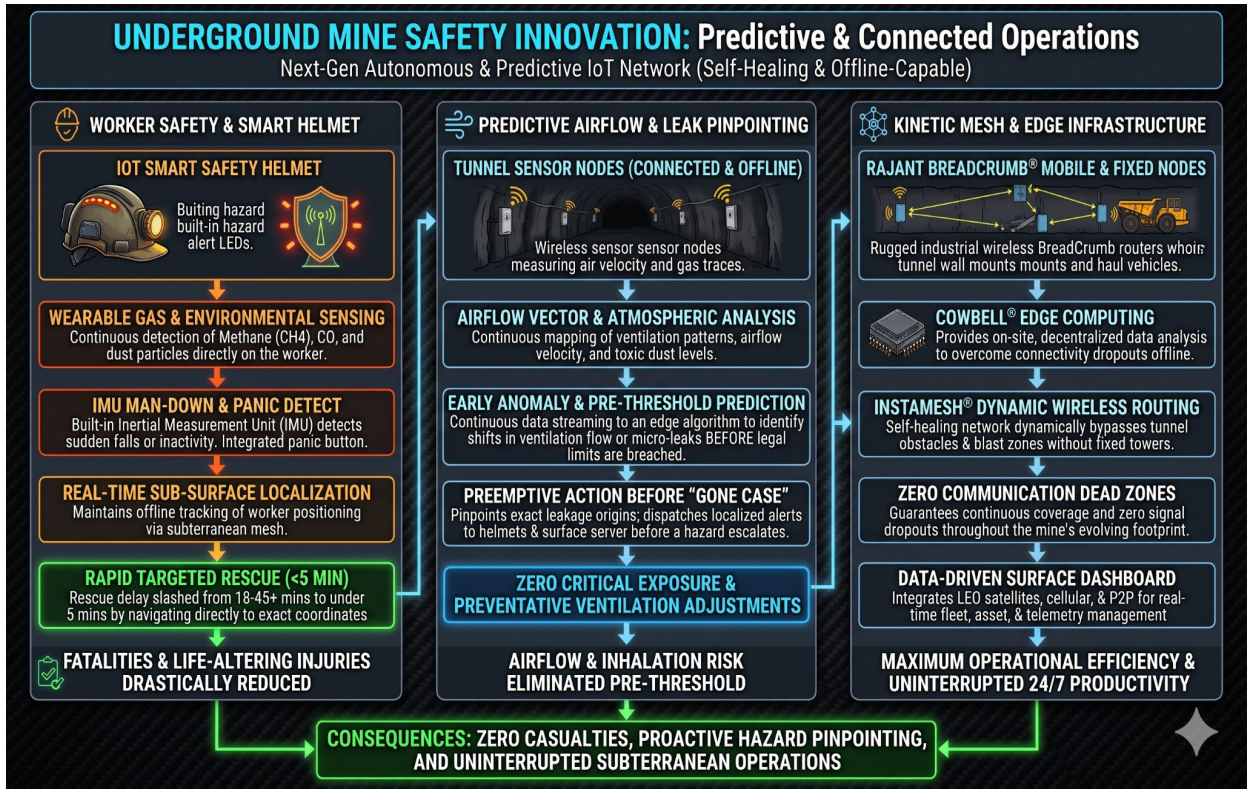
Target Beneficiary & Before-State User Flow

- **Target User:** Underground miners, ventilation officers, surface environmental monitoring teams, and emergency rescue squads operating in deep subterranean mine galleries and surrounding catchment areas.
- **Operational Bottlenecks:** Intermittent and manual handheld gas checking; absence of automated fall or motion detection on traditional hard hats; fragmented communication networks with severe sub-surface blind spots; and manual, delayed water grab sampling at discharge outlets.
- **Before-State Flow:** Toxic gas and respirable dust build up undetected in unmonitored tunnels -> A lone worker collapses without an automatic panic trigger or man-down alert -> Shift leads discover the incident with a delay due to fragmented wired phones or radio dropouts -> Rescue teams search blind galleries without real-time tracking coordinates -> In parallel, toxic mine drainage discharges unchecked into local waterways for days or weeks before manual lab results return.
- **Systemic Consequences:** Delayed evacuations, fatal poisoning or chronic respiratory illnesses; extended rescue delays during golden-hour emergencies; and widespread downstream environmental contamination triggering severe regulatory compliance penalties.

User Journey

GUIDELINE SLIDE

Visual "After State" Requirement



Target Beneficiary & Proposed Future User Flow

- **Target User:** Underground mine workers (such as face miners, loaders, and drillers), subsurface technical staff, and mine rescue teams operating in subterranean tunnels, galleries, and deep shaft systems.
- **Operational Bottleneck:** Undetected toxic gas & dust hazards due to manual or fragmented monitoring, along with slow and blind emergency search operations because there is no way to locate workers underground when communication drops or incident occur.
- **Proposed Future-State Flow:** Wireless sensor nodes & smart helmets continuously monitor the environment-> Micro-anomalies (like shift in airflow direction or trace gas ppm changes) are baseline-tracked by an edge-computing prediction algorithm->Gas/dust build-ups are pinpointed at their source before hitting thresholds-> Instant, direct wireless hazard alerts flash on smart helmets-> In case of a fallback, IMU man-down detection or the manual panic button instantly broadcasts precise offline sub-surface localization-> Rescue teams navigate straight to the exact worker spot within < 5 mins.
- **Sector Constraints & Capabilities:** Heavy subterranean terrain blocking cellular, LTE, and GPS is bypassed using an offline-capable, self-healing wireless mesh network. Relies entirely on low-cost wearable smart helmets and distributed non-line-of-sight nodes, preserving uninterrupted tracking and gas detection entirely without continuous surface fiber or cellular uplinks.

Example: Problem Context & User Journey

PROJECT TERRAWEED EXAMPLE

₹4,200/Ac	62 Hrs/Ac	28% Loss	68 dB(A)
MANUAL WEEDING EXPENSE	MANUAL LABOR TIME	YIELD LOSS VIA WEED STUNT	SAFETY EXPOSURE LIMIT



Figure: Vidarbha smallholder inspecting severe weed infestation along 45cm cotton rows

Target Beneficiary & Before-State User Flow

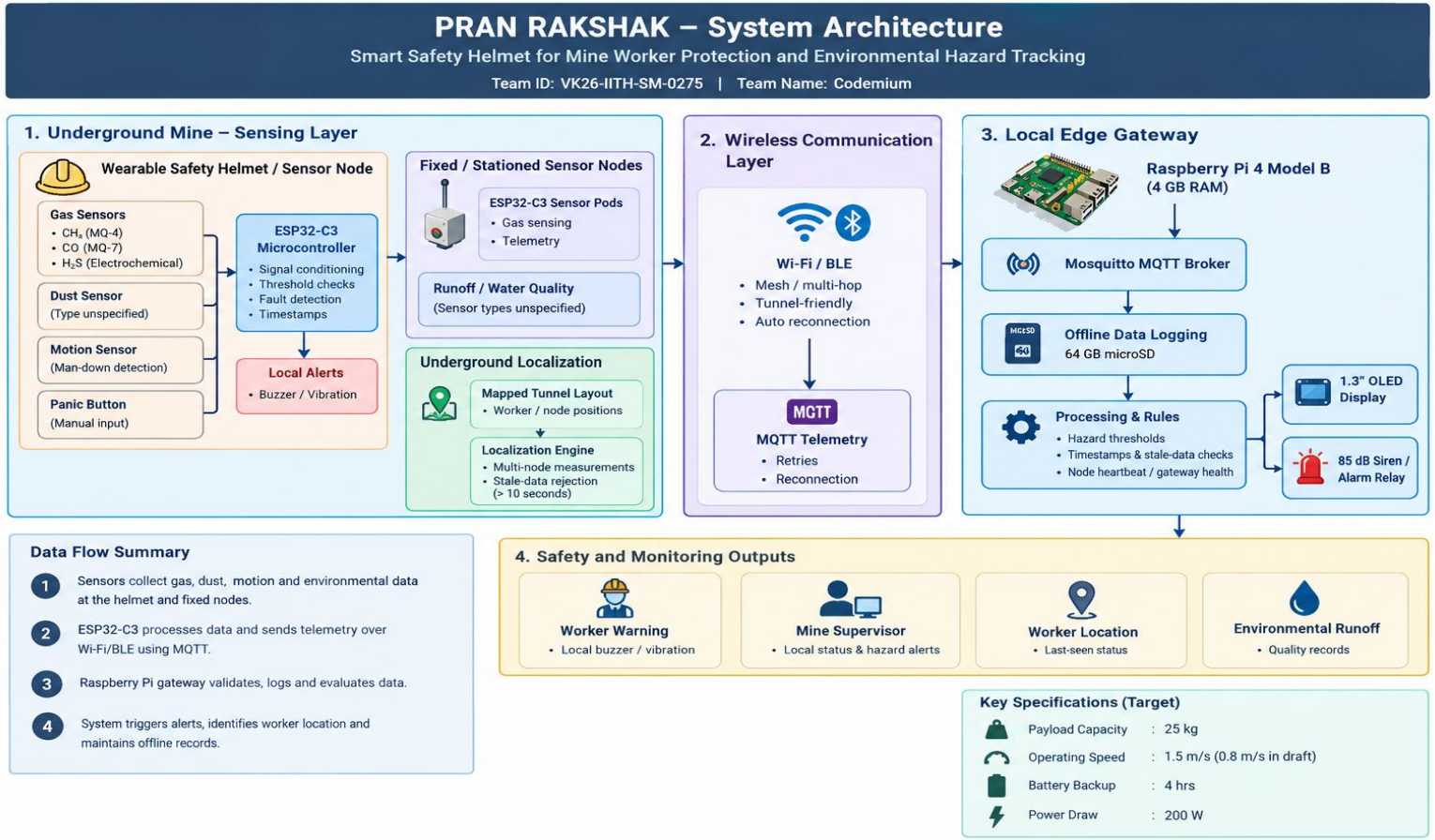
- **Target User:** Smallholder cotton farmers in Vidarbha, Maharashtra operating 1–4 acre farms with 45cm crop row spacing.
- **Operational Bottleneck:** Severe seasonal labor shortages during the 20-day weed-competition window force reliance on manual sickle scraping or drift-heavy herbicides.
- **Before-State Flow:** Weed infestation emerges (Day 1-5) → 72-hr delay finding scarce farm labor → Slow manual hoeing (0.15 ac/day) causing back strain → Late glyphosate spraying damages 14% cotton plants.
- **Public Search Disclosure:** Commercial weeding tractors require >30 HP and >1.2m row spacing, crushing crops. Hand-carried weeders cause severe musculoskeletal fatigue after 3 hrs.

Exemplar Submission Note: Team ID: VK-2026-AG-108 | Track: Smart Agriculture & Mechanization | Core Innovation: Compact autonomous active-tine row weeder.

System Architecture & Specs

GUIDELINE SLIDE

System Block Diagram



Target Specifications

PRAN RAKSHAK targets under 10 s end to end gas alert latency (**threshold crossing to helmet vibration and OLED warning**) at 1 Hz sampling of **CH4, CO and H2S**. Each wearable node is designed for 8 hours of battery backup (one full shift) on a **3.7 V 2500 mAh LiPo**, using duty-cycled sensor heaters to keep average draw near 1 W. The IP67-targeted pod weighs about 120 g (52 × 88 × 28 mm). **Nodes report to a Raspberry Pi 4 gateway over Wi-Fi/MQTT using kinetic mesh**, and the gateway runs fully offline with at least 7 days of **local logging**. A lost node is **flagged within 30s**, man-down triggers after 30 s of no motion, and **workers are localized to the nearest node zone.**

Hardware Selection



Deploy Sensor Nodes

- Wearable sensor node integrated into the worker's helmet
- Fixed sensor nodes installed inside the mine
- Wearable node contains MQ-4 methane, MQ-7 carbon monoxide, electrochemical H_2S, dust, motion, and panic button sensors
- Fixed nodes monitor CH_4, CO, H_2S, runoff, and water quality

Collect and Process Data

- ESP32-C3 microcontroller for data collection, signal conditioning, threshold checking, fault detection, timestamping, and telemetry

Transmit Wireless Data

- Communicate through Wi-Fi and Bluetooth Low Energy (BLE)
- Use mesh or multi-hop communication to transmit readings and alerts

Manage Edge Gateway

- Raspberry Pi 4 Model B with 4 GB RAM and Mosquitto MQTT broker
- 64 GB microSD card for offline data logging
- Process data using hazard threshold detection, timestamp validation, stale-data rejection, and health monitoring

Output Safety Alerts

- 1.3-inch OLED display for real-time status and alerts
- 85 dB siren for audible hazard warnings
- Mine supervisor interface for notifications
- Worker localization module using tunnel map

Example: System Architecture & Specs

PROJECT TERRAWEED EXAMPLE



Engineering Target Specifications & Subsystems

Parameter	Target Spec	Selected Hardware / Protocol
Gas sensing	CH4, CO, H2S at 1 Hz sampling	MQ-4 (CH4), MQ-7 (CO), H2S electrochemical sensor, analog conditioning into the ESP32-C3 ADC
Alert latency	<10 s, threshold crossing to helmet alert	On-node threshold check; helmet vibration motor and OLED
Node controller	160 MHz, deep sleep between samples	ESP32-C3-WROOM-02-N4 (RISC-V, Wi-Fi/BLE)
Battery backup	8 hrs (one shift), heaters duty-cycled	3.7 V 2500 mAh LiPo, TP4056 charge and protection, low-dropout 3.3 V regulator
Link and protocol	Lost node flagged in 30 s, auto reconnect	Wi-Fi + MQTT (QoS 1), topics, `sensors/<node_id>/telemetry`, `/status`, `gateway/alerts`
Gateway	Fully offline, 7+ days local logging	Raspberry Pi 4 (4 GB), Mosquitto broker, threshold rules, local dashboard, 64 GB microSD

Architecture Rationale: ESP32-C3 with Wi-Fi and MQTT was chosen for low-cost, offline-capable communication between sensor nodes and the Raspberry Pi gateway, enabling real-time hazard alerts and worker monitoring.

Node & Gateway Architecture Specs

GUIDELINE SLIDE

- **ESP32-C3 Sensor Nodes:** Wearable & stationed pods with ESP32-C3 (RISC-V, Wi-Fi/BLE) for low-power telemetry.
- **Hazardous Gas Sensing:** Integrate calibrated CH4, CO, and H2S sensors with intrinsic safety and analog preconditioning.
- **Raspberry Pi Server:** Deploy local Raspberry Pi base station for offline data logging, MQTT brokering, and alerts.



Example: ESP32-C3 Node Hardware & Enclosure

PROJECT TERRAWEED EXAMPLE



Figure: CAD assembly of wearable ESP32-C3 sensor pod with gas diffusion manifold & IP67 seals

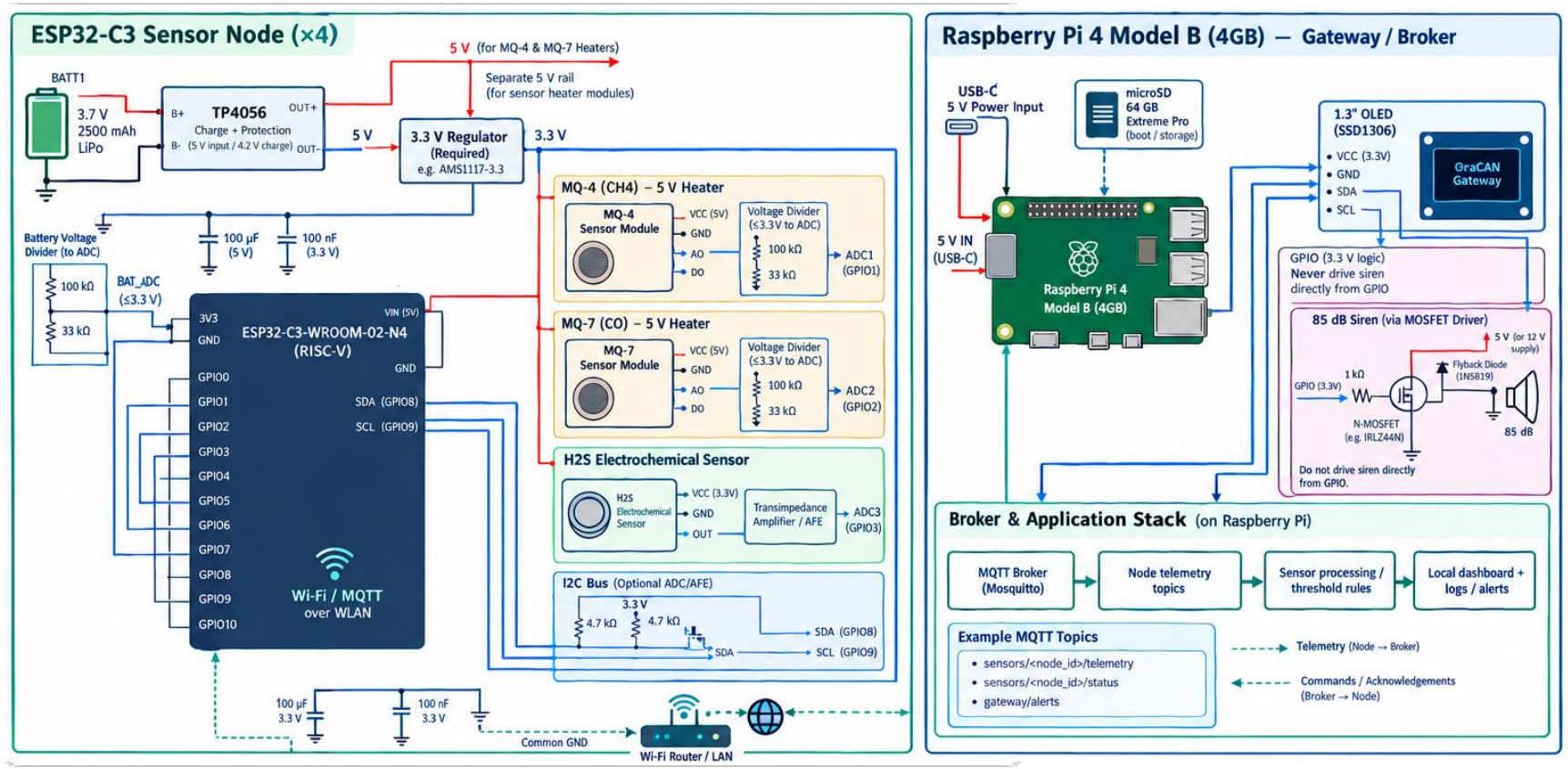
Hardware Specifications & Enclosure Design

- **Enclosure Material:** Polycarbonate-ABS blend (UL94 V-0 flame-retardant) with silicone gasket; IP67 ingress rated against toxic slurry & mine dust.
- **Core Processing Unit:** ESP32-C3 RISC-V 32-bit MCU @ 160MHz; hardware crypto accelerator, low deep-sleep draw, and native 2.4 GHz Wi-Fi/BLE 5 mesh.
- **Gas Intake & Filtration:** Sintered brass/stainless steel flame arrestor mesh over MQ-4 (CH4), MQ-7 (CO), and H2S chambers preventing thermal spark ignition.
- **Power Architecture:** 3.7V 2500mAh Li-Po battery with intrinsic safety fuse, current limiting, and low-quiescent LDO; achieves >16 hours per shift.
- **Impact Integrity:** Drop-tested for 2.0m shock impacts onto granite; shock-absorbing silicone dampeners isolate internal sensor PCB and antenna.

Enclosure Optimization: Vortex gas diffusion port reduces response latency from 8.2s to 2.1s while deflecting ambient airborne coal particulates.

Electronics Schematics & BOM Details

GUIDELINE SLIDE



Circuit Schematics: ESP32-C3 pinouts, ADC gas sensor conditioning, 3.3V LDO regulation, and TP4056 charging.

Gateway Architecture: Detailed Raspberry Pi server wiring, MQTT broker network interfaces, alarm relays, and 5V/3A supply.

Itemized BOM Table: Listed all parts, exact MPNs, quantities per node, unit costs, and domestic vendor sourcing lead times.

Example: Electronics Schematics & BOM

PROJECT TERRAWEED EXAMPLE

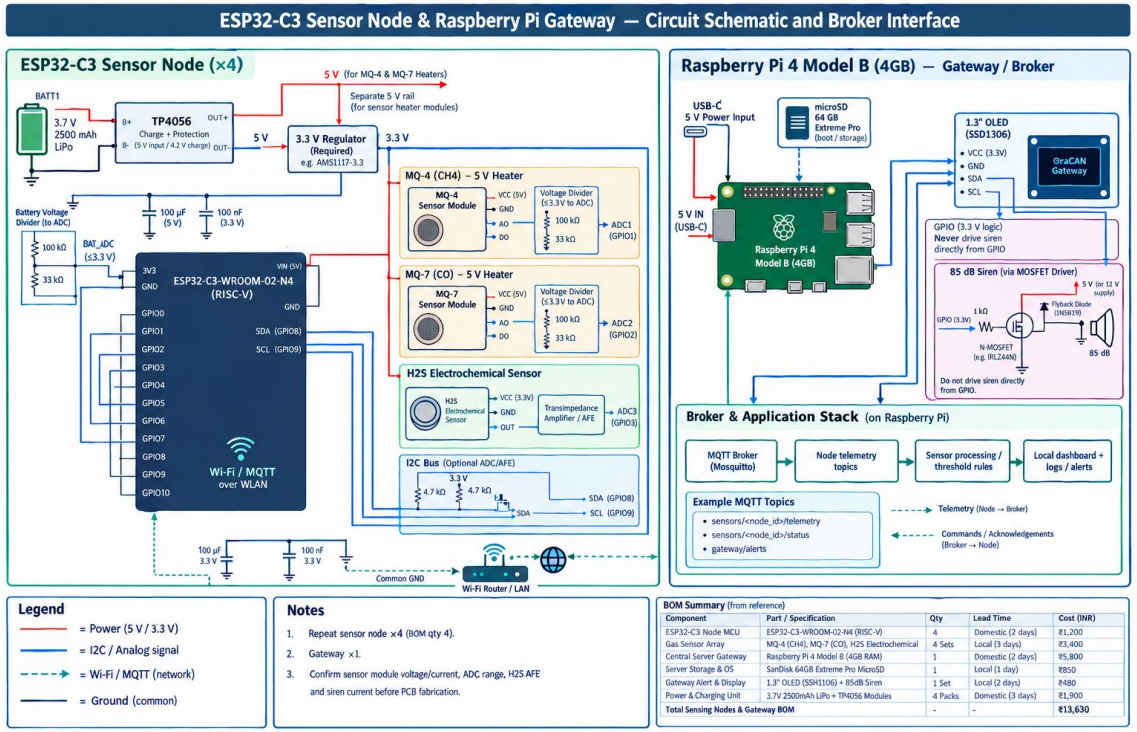


Figure: Circuit schematic of ESP32-C3 sensor node & Raspberry Pi gateway broker interface

Itemized Prototype Bill of Materials (BOM)

Component	Part / Specification	Qty	Lead Time	Cost (INR)
ESP32-C3 Node MCU	ESP32-C3-WROOM-02-N4 (RISC-V)	4	Domestic (2 days)	₹1,200
Gas Sensor Array	MQ-4 (CH4), MQ-7 (CO), H2S Electrochemical	4 Sets	Local (3 days)	₹3,400
Central Server Gateway	Raspberry Pi 4 Model B (4GB RAM)	1	Domestic (2 days)	₹5,800
Server Storage & OS	SanDisk 64GB Extreme Pro MicroSD	1	Local (1 day)	₹850
Gateway Alert & Display	1.3" OLED (SSH1106) + 85dB Siren	1 Set	Local (2 days)	₹480
Power & Charging Unit	3.7V 2500mAh LiPo + TP4056 Modules	4 Packs	Domestic (3 days)	₹1,900
Total Sensing Nodes & Gateway BOM	**₹13,630**	**-**	**-**	**-**

Procurement Status: All ESP32-C3 modules, gas sensors, and Raspberry Pi gateway hardware sourced domestically with standard lead time under 7 days.

Your Maker Journey (1-2 Slides)

GUIDELINE SLIDE

Authenticity Verification

Proof of personal fabrication:

- Include high-resolution lab photos of team members actively modeling, soldering, and assembling.
- Team members must wear institutional ID badges in the frame.
- If a faculty mentor is attached, include a joint lab review photo.

Work-In-Progress Software Evidence

Screenshots of original design assets:

- CAD software UI showing active parametric feature tree with unsuppressed sketches.
- EDA layout software (KiCad / Altium) with DRC rule-check logs.
- FEA convergence plots and mesh statistics.

Raw Repository Linkage

Provide an accessible Google Drive / GitHub repo link:

- Must include raw native files (.STEP, .SLDPRT, .kicad_sch, .m).
- File creation timestamps will be audited to confirm progress during Stage 2.
- No embedded videos; provide external links only.

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Example: Maker Journey

PROJECT TERRAWEED EXAMPLE



Figure: Team members conducting motor dyno & telemetry testing at COEP Pune Maker Lab

Stage 1 to Stage 2 Engineering Iterations

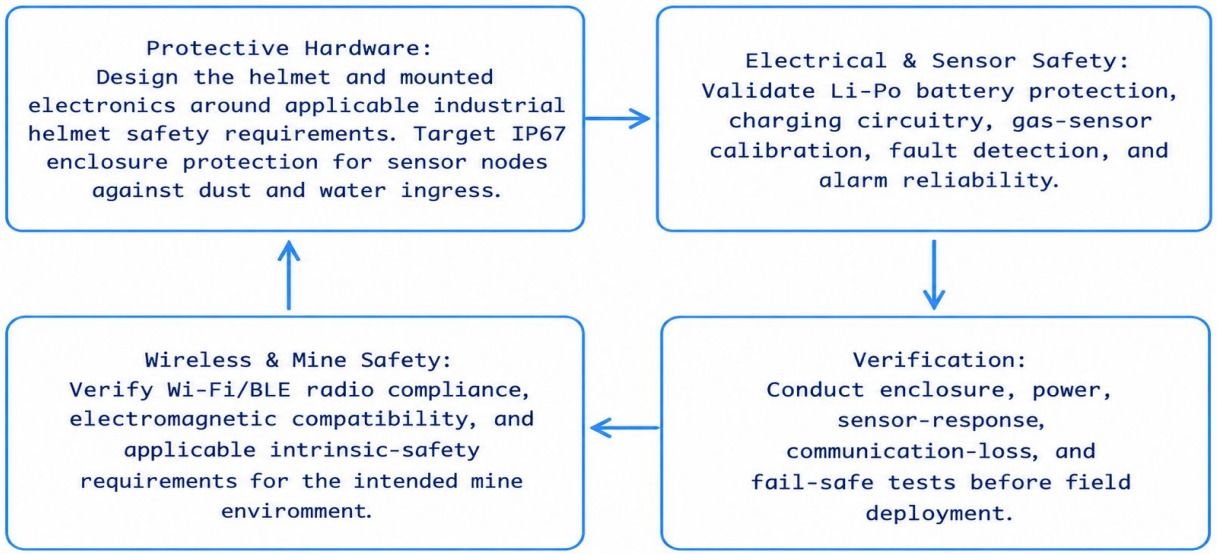
- **Iterative Pivot:** Replaced passive cultivator shank from Stage 1 with dual active counter-rotating discs after torque calculations indicated 40% soil clogging in clayey black cotton soil.
- **CAD Milestone:** 47 discrete parametric parts modeled in SolidWorks 2024; completed kinematic multi-body simulation of 4-bar steering on Aug 24, 2026.
- **Electrical Testing:** Validated current surge on 24V bus during tiller stall (peaked at 18.2A for 120ms; managed safely by flyback diodes and smart BMS cutoff).

- **Verified Drive Repository:** drive.google.com/drive/folders/terraweed-stage2-audit (Contains timestamped .SLDASM, BOM.xlsx, KiCad v8 schematics, and raw FEA stress logs).

Institutional Sign-Off: Work audited and certified by Faculty Guide Prof. S. K. Kulkarni, Dept. of Mechatronics, COEP Technological University Pune.

Safety Mitigation

Standards Compliance



Potential failure mode	Risk / effect	Embedded software protection
Gas sensor drift or failure	Incorrect hazard readings	Calibration checks, threshold validation, sensor fault alerts
Power drop or low battery	Sensor or node shutdown	Battery monitoring, low-power warning, controlled shutdown
Wireless communication loss	Delayed hazard alerts	Heartbeat monitoring, automatic reconnection, local alarms
Sensor data delay	Late hazard detection	Timestamp validation, stale-data detection, watchdog timers
False or missed alarms	Unsafe worker response	Threshold validation, filtering, alarm self-tests
Gateway/server failure	Loss of central monitoring	Local node level alerts, offline data logging, gateway health monitoring

Hazard Mitigation



Risk analysis

Assess gas-sensor faults, missed alerts, power loss, communication failure and false alarms using the nodes based on the ventilation blueprint.

Fail-safe response

Keep local alarms active and log critical events without internet using slave/master controllers supporting emergency response.

Failure detection

Monitor sensor health, low battery and disconnected nodes; flag unreliable readings.

Instant local alerts

Trigger on helmet's OLED and vibration alerts when validated threshold limits are reached to the master controller.

Integrated Ventilation Safety Solution

Ventilation safety:
Combining airflow monitoring with wireless connectivity for rapid alerts & data driven ventilation management.

Airflow detection:
Real time airflow sensing identifies changing ventilation conditions, enabling faster detection of inadequate airflow and supporting proactive mine safety

Real-time data processing:
Cowbell edge computing supports local data processing for responsive monitoring applications.

Kinetic Mesh Connectivity:
Multi-radio, multi-path wireless networking maintains communication between mobiles, environmental sensors & monitoring systems across the mine.

Continuous coverage:
Nodes & mesh routing adapt to changing site conditions, helping maintain connectivity as equipment & operations move.

Integrated safety alerts:
Connecting airflow sensors to the mesh network enables timely transmission of abnormal condition alerts to operators & control rooms.

Made with Napkin

Roadmap

Design FMEA (Failure Mode and Effects Analysis)

Failure Mode	Potential Hazard	Embedded Fail-Safe / Mitigation Response
Tiller Stone JamGas sensor drift or dead sensor (MQ-4 / MQ-7 / H2S)	False “safe” reading, so a worker stays in a methane/CO/H2S zone	Baseline-drift check at every boot; reading stuck or flat-lined for >60 s flags the sensor faulty; node raises a “sensor fault” alert and never shows “safe”
Node loses link to gateway (tunnel bends/branches)	Worker unlocatable, alerts not delivered	Local threshold check on the ESP32-C3 with buzzer/vibration alert, no server needed; auto-reconnect plus MQTT retries; gateway flags the node “lost” if last-seen >30 s
Stale or clock-skewed readings	Wrong localization, wrong leak source	Every packet timestamped; gateway discards readings older than 10 s before running localization

Fabrication Phase (Months 1 - 2)

Oct 2026: Finalize the schematic and run the KiCad DRC. Order the PCB and components. Bench-calibrate the MQ-4/MQ-7/H2S sensors against a reference gas.

Nov 2026: Assemble nodes and helmet enclosure. Build the Pi 4 gateway, the Mosquitto broker and the threshold rules. Bring up the OLED and siren.

Integration & Field Demo (Months 3 - 4)

Dec 2026: Test mesh and link-loss behavior. Tune man-down detection. Run the watchdog and brownout tests. Run a first full multi-node localization test on a mapped corridor.

Jan 2027: Run a long-duration soak test (battery life, drift). Final tuning and a rehearsed live demo for Stage 3.

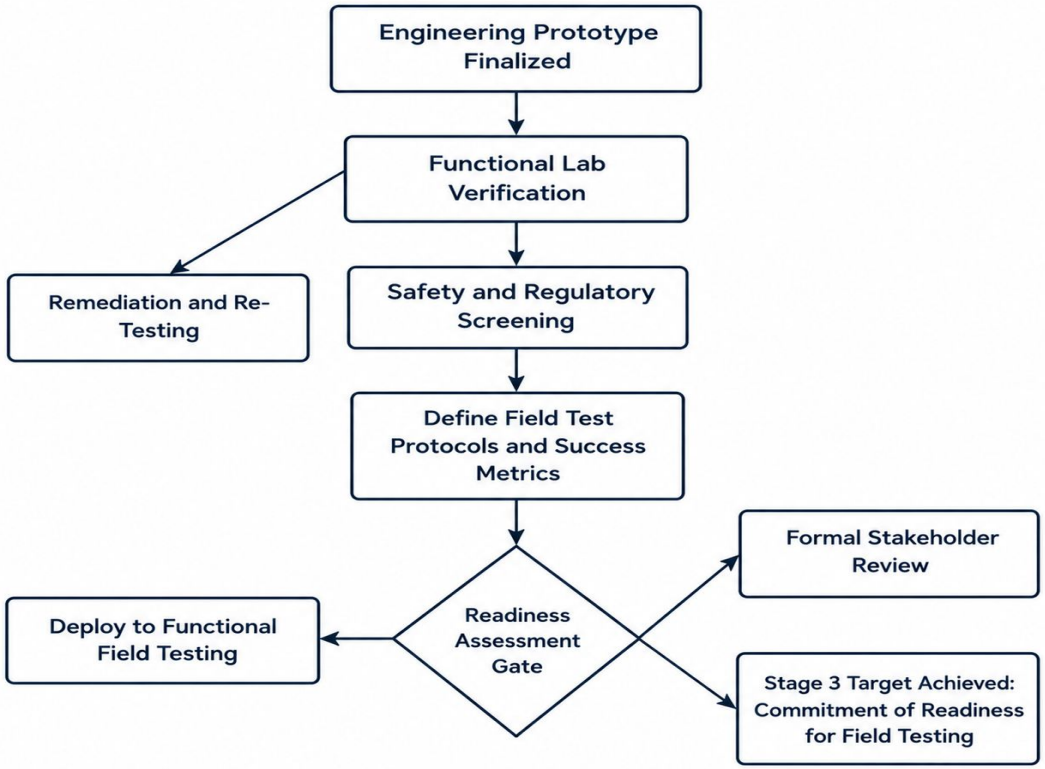
Roadmap

GUIDELINE SLIDE

4-Month Fabrication Plan

Fabrication Phase (Months 1-2):
The first two months deliver a working hardware. We finalize the schematic, **run the KiCad DRC** and **order the PCB**, then **calibrate the MQ-4, MQ-7 and H2S sensors** against a reference gas so every reading has a known error. We then **assemble four sensor nodes, the helmet unit and the Raspberry Pi 4 gateway**, and **bring up the MQTT broker, threshold rules, OLED**. Exit criterion is **every node sends timestamped telemetry to the gateway**, and a gas threshold crossing **triggers a local alarm with even minimal internet.**

Integration & Field Demo (Months 3-4):
The last two months turn the working hardware into a system. We **test mesh behavior** and **kinetic mesh connectivity**, and run the watchdog and brownout tests. We then **run a multi node localization test on a mapped corridor using the ventilation blueprint** and a long duration test for battery life and sensor drift. Stage 3 in which the **gas alert, panic button, man alert and node loss flagging all work as per the stated targets.**



Technical Challenges & Risks

Sensor accuracy and Calibration

Sensor can be affected by the surrounding environment and can produce less accurate outputs and respond slowly

Poor **placement of the sensors** may cause issues and cause detection problem

Airflow Direction and Velocity

As we are using the direction of air to find the source of leak, there can be many situations where the flow changes and we predict wrong.

Synchronization/Server delay

Nodes may have different clock offsets, while gas sensors can take time to respond to concentration changes.

Network Reliability

The network may be interrupted continuously due to bends and branches of caves.

Power Limitations

ESP32-C3 memory, GPIO and ADC limitations; sensor power requirements; unstable supplies; and potential overheating or resets.

Potential Solutions & Risk Mitigation (1 Slide)

GUIDELINE SLIDE

Potential Solutions

To resolve environmental and spatial variables, the system integrates routine multi-point calibration, digital noise filtering, and temperature-humidity compensation to prevent drift and false readings. Prior 3D tunnel mapping ensures fixed nodes are installed at calculated intervals and stratifications, backed by smart helmets that continuously monitor workers in active drifts. To manage shifting airflow, tunnel nodes continuously track ventilation velocity and directional vectors, allowing the edge algorithm to dynamically adjust, pinpoint leak origins, and dispatch direct helmet warnings well before gas limits are breached.

To address networking and edge hardware bottlenecks, the platform utilizes a self-healing peer-to-peer mesh architecture (such as Rajant Kinetic Mesh with BreadCrumb nodes) that routes around subterranean obstacles without fixed-tower dependencies. Localization and predictive anomaly models run directly on local edge compute modules, ensuring real-time alerts even when cut off from surface links. Additionally, strict capture-timestamping prevents server latency errors, while hardware watchdogs and regulated power circuits safeguard edge microcontrollers against memory crashes and brownouts.

Risk Mitigation

To mitigate sensor failures and environmental interference, the architecture uses multi-tiered hardware and algorithmic fallbacks. If primary optical gas sensors drift or fail, the system immediately switches to wide-spectrum catalytic or metal-oxide backup sensors to maintain threshold alarms. If an anemometer clogs, the algorithm reverts to historical ventilation baselines to estimate gas dispersion. Additionally, all modules are protected by dual-stage sintered metal and hydrophobic PTFE membranes against moisture, slurry, and dust fouling.

To handle communication dropouts and power faults, redundant transmission layers and supervisory controls are deployed. If cave-ins block primary Kinetic Mesh channels, nodes fall back to low-frequency sub-GHz protocols to transmit distress coordinates, while non-volatile local buffers cache unsent telemetry until links reconnect. At the hardware level, external watchdog circuits trigger automatic resets during system hangs or brownouts, and onboard backup batteries activate low-power beacon modes to keep worker localization active during outages.

Technical Challenges & Risks

Sensor Calibration Problems

How it can be solved?

Calibrate each sensors properly
Apply filtering and temperature compensation where supported

Improper placement of Sensors

How it can be solved?

Proper mapping of the tunnel is required before placement
Place sensors at suitable intervals.

Airflow Direction and Velocity

We have to regularly update the localization model when ventilation changes
Measure airflow direction regularly

Server Delay

Synchronize node clocks timestamp every reading.
Detect stale or delayed readings before localization.

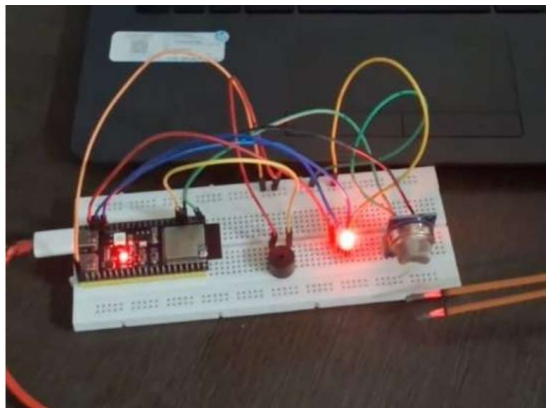
Network Delay

Implement automatic reconnection and message retries.
Monitor node connectivity and last-seen timestamps.

Power Limitations

Add watchdogs, brownout handling and reset diagnostics.
Use a stable, properly rated power supply.

Image Sources



https://www.electronicwings.com/users/KhushaliVadaliya/projects/5953/smart-gas-leak-detector-using-esp32-and-blynk

We can see which sensors are going to be used with our microcontroller.

The diagram shows a rough plan where the sensors are going to be placed in the tunnel

https://pmc.ncbi.nlm.nih.gov/articles/PMC11902780/?

Figure 9.

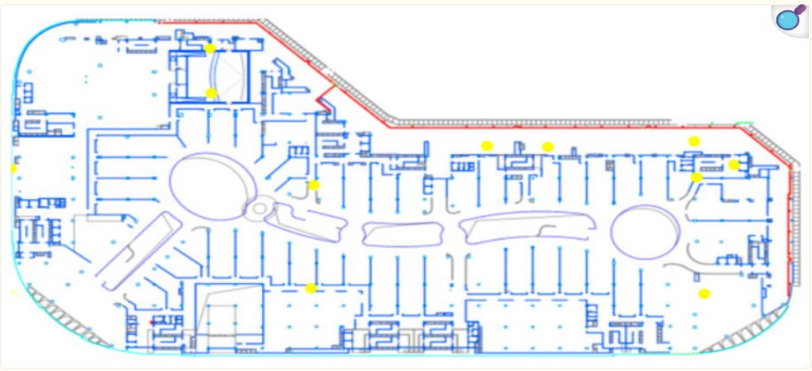
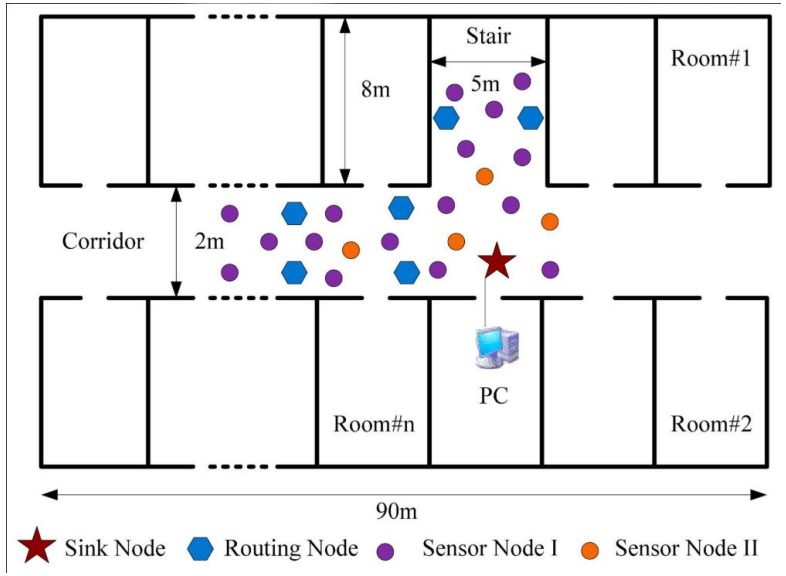


Image Sources



Useful for understanding how different nodes communicate with each other and with the sensors

https://pmc.ncbi.nlm.nih.gov/articles/PMC3348832/

References & Sources

GUIDELINE SLIDE



https://www.cdc.gov/niosh/media/pdfs/2026/03/2010-141.pdf

Source: National Institute for Occupational Safety and Health (NIOSH)



https://archive.cdc.gov/www_cdc_gov/niosh/mining/topics/atmosphericmonitoring.html

Source:https://www.cdc.gov/

https://stacks.cdc.gov/view/cdc/230545?utm_source=chatgpt.com

Source:U.S.centers for disease control and prevention